Cattle on the urban edge: the multiple benefits of extensive grazing on a Mediterranean-climate working landscape

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Livestock are increasingly scrutinized for their climate cost, yet in Mediterranean-climate regions extensive cattle grazing occupies working landscapes whose broader value is rarely assessed as a whole. We integrate published field evidence for the rangelands of Yolo County, California, around the university town of Davis, and quantify five benefits of continued grazing: reduced wildfire fuel and fire behavior, maintenance of native grassland biodiversity, soil-carbon storage, forage-based food and economic output, and retention of open space against conversion. We weigh these against the enteric methane that makes beef the most greenhouse-gas-intensive common protein. A scenario comparison of continued grazing, land abandonment, and urban development shows that grazing retains the widest set of benefits, and that the principal threat to them is irreversible conversion rather than grazing itself. The results argue for valuing peri-urban grazing lands as multifunctional infrastructure.

Cattle grazing is the dominant use of rangelands worldwide, and in California extensive grazing covers roughly 23.1 million hectares of potential grazing land, about 60 % of which was grazed in the mid-2000s[1]. These lands sit disproportionately on the expanding edges of cities, where the same ground is contested by housing, intensive agriculture, and abandonment. The public conversation about cattle has narrowed to a single axis, greenhouse-gas emissions, on which ruminant systems perform poorly: enteric fermentation is the largest single source of livestock emissions globally[2]. Framed only this way, the rational policy is fewer cattle and less grazed land. Yet the land does not disappear when the cattle leave; it becomes something else, and the relevant comparison is not grazing against a blank slate but grazing against the alternative uses that replace it.

On any given acre, a working landscape delivers more than one service at a time, and these services are seldom measured together. Targeted grazing removes the fine fuels that carry grassland fire[3,4]. Grazing at moderate intensity sustains the native plants and grassland birds of California's annual grasslands, which are otherwise lost to a dense thatch of invasive annuals[5,6]. Grazed rangelands store large stocks of soil carbon and, under compost amendment, can draw down additional greenhouse gases[7,8]. The same acres support a ranching economy and pollination services worth hundreds of millions of dollars[9,10]. Each of these findings is well established in isolation; what is missing is an integrated account that lets them be weighed against one another and against the methane cost, at the scale of a real place.

Studying one such place, we assemble the published evidence for the rangelands of Yolo County surrounding Davis, California, and quantify five benefits of continued extensive grazing alongside its principal cost. We then compare three futures for the same land (continued grazing, abandonment, and development) across all five dimensions. Our aim is not to declare cattle good or bad in the abstract, but to show what a peri-urban Mediterranean-climate community gains and loses under each choice, and to identify where the largest and least reversible effects lie.

Results

A working landscape on the urban edge

Much of Yolo County remains agricultural: it contains 478 555 acres of land in farms, of which 164 896 acres (about a third) are pastureland and a further 22 690 acres are woodland, supporting on the order of 53 000 cattle and calves[1]. This composition (Fig. 1a) is typical of the lower Central Valley edge: a matrix of cropland, annual grassland grazed by beef cattle, and oak woodland, abutting a growing university town. At the state scale, grazed rangeland is extensive but under pressure (Fig. 1b), and the benefits that follow all flow from keeping this land in grazed open space rather than converting it (Fig. 1c).

Grazing lowers fine-fuel loads and modeled fire behavior

On grazed range, livestock remove the standing herbaceous biomass that carries surface fire. In Mediterranean grasslands, targeted grazing has been shown to remove between 71 and 83 % of fine fuels, with associated reductions in litter depth and vegetation height[4]. Represented as a grazed-versus-ungrazed contrast in fine-fuel load (Fig. 2a), this shifts a site from the high-biomass state that supports fast-spreading, high-intensity fire toward loads below the thresholds at which extreme fire behavior is modeled[3]. In fire-behavior modeling of herbaceous fuels, fireline intensity and rate of spread rise steeply with fuel load, and grazing moves a site down that curve, below the intensity at which direct suppression becomes unsafe (Fig. 2b). The benefit is conditional, and honestly so: under extreme fire weather (low fuel moisture and high winds), moderate grazing has limited effect, because weather rather than fuel then governs fire behavior[3] (Fig. 2c). The value of grazing for fire is therefore greatest in grasslands and under the moderate conditions that prevail for most of the fire season.

Moderate grazing maintains grassland biodiversity

Shaped by a long history of grazing, California's annual grasslands accumulate a productive but species-poor thatch of invasive annual grasses that suppresses native plants once grazing is withdrawn. In a study spanning grazed and ungrazed grasslands, livestock grazing increased native plant cover, and native cover in turn was a leading predictor of the abundance of three grassland bird species[5] (Fig. 3a). A multi-year removal experiment found that ungrazed plots lost native species while grazed plots did not: over five years, ungrazed plots lost 4.5 native species on average while grazed plots gained 0.5[6] (Fig. 3b,c). Grazing also stabilized native plant cover and diversity from year to year. The available evidence is a grazed-versus-ungrazed contrast rather than a full grazing-intensity gradient, so the finding supports moderate grazing as a maintenance tool for native diversity, not a claim that heavier grazing

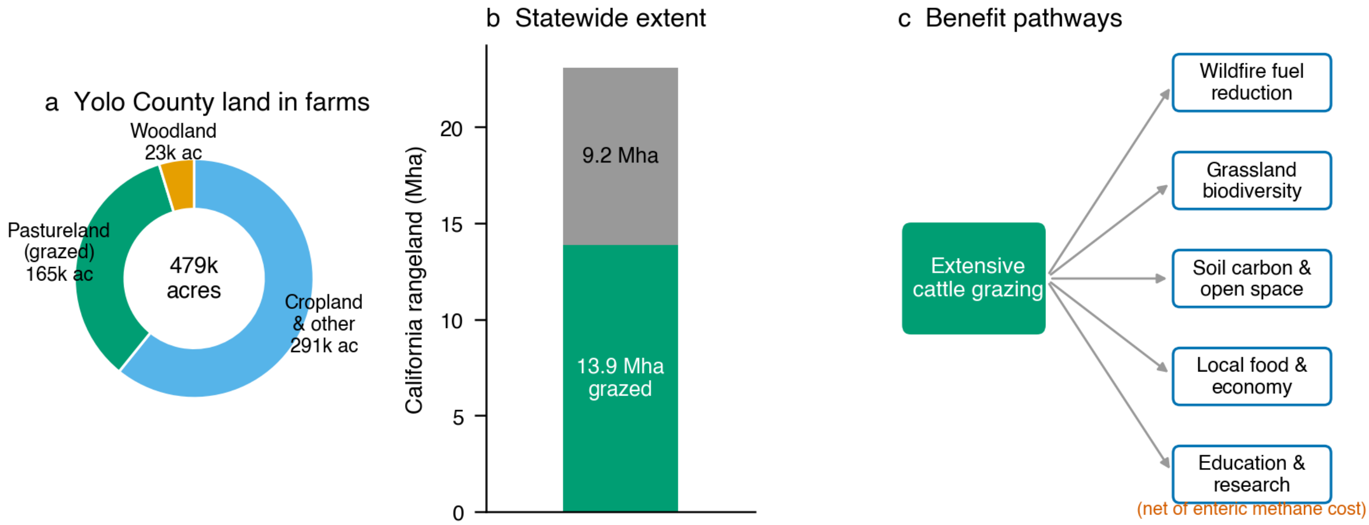

Figure 1. Study system and benefit pathways. **a**, Yolo County land in farms by use (2022 Census of Agriculture): pastureland and woodland together account for about 39 % of farmland. **b**, California rangeland extent, of which roughly 60 % is grazed. **c**, The five benefit pathways of extensive grazing evaluated here, each net of the enteric-methane cost. Land-use values are from the USDA Census of Agriculture and Cameron et al.[1]

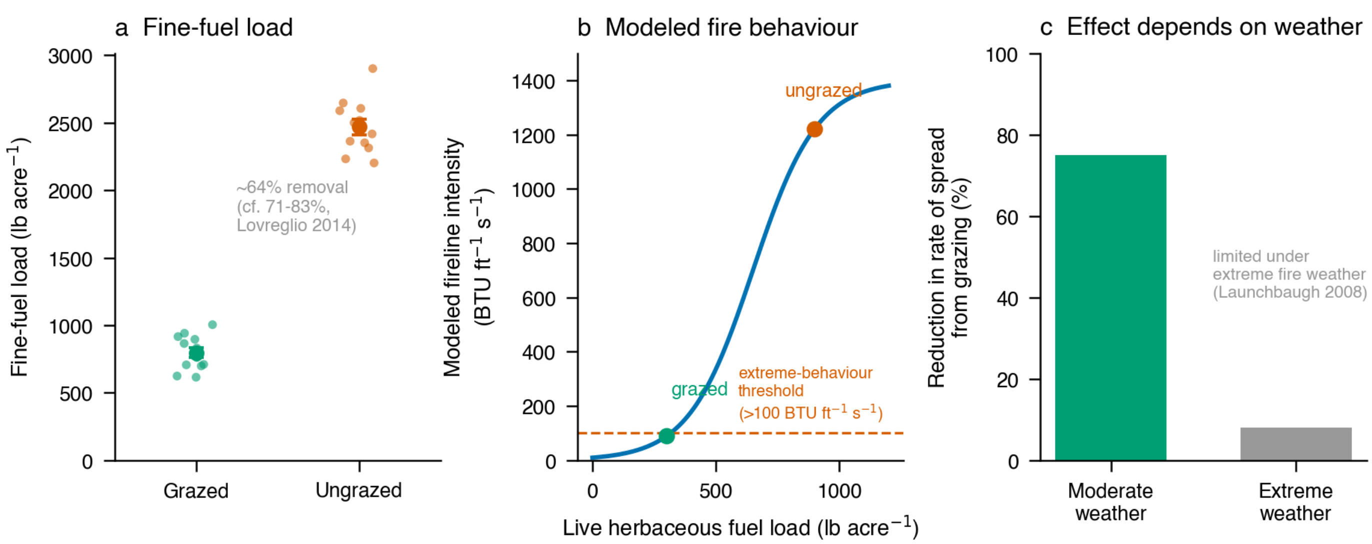

Figure 2. Grazing reduces fine-fuel load and modeled fire behavior. **a**, Fine-fuel load under grazed and ungrazed conditions; points are illustrative, calibrated to the 71–83 % fuel removal reported by Lovreglio et al.[4] (mean ± 1 s.e., $n = 12$). **b**, Modeled fireline intensity as a function of live herbaceous fuel load, reproducing the monotone relationship and the 100 BTU ft^{-1} s^{-1} extreme-behavior threshold of Launchbaugh et al.[3]; grazed and ungrazed fuel loads marked. **c**, Reduction in rate of spread from grazing under moderate versus extreme fire weather, illustrating the weather dependence[3,4].

is better.

Soil-carbon gains and the greenhouse-gas balance

Carbon can accumulate in substantial amounts in the soils of grazed and organically managed working lands. At the Century Experiment on Russell Ranch in Yolo County itself, an organically managed system receiving composted manure and winter cover crops gained 21.8 megagrams of carbon per hectare over the full two-meter soil profile in nineteen years, about 1.09 megagrams of carbon per hectare per year[7] (Fig. 4a); conventional and cover-crop-only systems did not accrue deep-profile carbon. A single application of compost to grazed annual grassland has been estimated to yield a net greenhouse-gas benefit of about 22.6 megagrams of carbon-dioxide equivalent per hectare over three years, driven mainly by emissions avoided when organic waste is diverted from landfill, with a smaller direct soil sink[8] (Fig. 4b). Integrated crop–livestock grazing has likewise been found to raise subsoil carbon[11].

Livestock are scrutinized for a reason, and these gains must be set against it. Enteric fermentation is the largest single source of livestock greenhouse-gas emissions, and cattle account for the majority of them; ruminant meat is the most emissions-intensive common protein, at 58 to more than 1000 kilograms of carbon-dioxide equivalent per kilogram of protein, against 3.7 for poultry[2] (Fig. 4c). Six to seven percent of the feed energy a grazing animal consumes is lost as methane[12]. The soil-carbon and waste-diversion benefits of well-managed grazing are real but bounded, and they do not by themselves cancel the methane cost; an honest account keeps both in view.

Forage, food, and the working-landscape economy

Used as pasture, the same rangeland produces food and supports a rural economy. Forage productivity and grazing capacity are highest on open grassland and decline through savanna to woodland, with grazing capacity of 6.8, 4.2, and 2.8 animal-unit-months per hectare respectively and corresponding forage use values of 120, 74, and 50 dollars per hectare[9] (Fig. 5a). At the scale of a typical 251-hectare ranch, the allocation of land between grassland and woodland trades annual grazing revenue against long-term carbon value, so that a balanced mosaic captures both to a degree that neither extreme does[9] (Fig. 5b). Beyond forage, California's rangelands support pollination services for roughly a third of the state's crops, valued at between 937 million and 2.4 billion dollars annually[10] (Fig. 5c). Ranching has been described as among the most ecologically sustainable segments of the domestic meat industry, precisely because it keeps large areas in low-input open space[10]. In a university town, these lands also carry an educational and research value, from the long-running experiments at Russell Ranch to teaching herds, that compounds their public return.

Integrating benefits across land-use futures

Shared among these benefits is a single vulnerability: they exist only while the land stays in grazed open space. Between 1984 and 2008, nearly 200 000 hectares of California rangeland were converted, about half of it to development[1], and integrated scenarios project a further loss of 22 to 37 % of grassland habitat and a doubling to tripling of developed land by 2100[13]. Comparing three futures for the same ground (continued grazing, abandonment, and development) across the five dimensions (Fig. 6a) shows continued grazing retaining the widest set of benefits, abandonment forfeiting fire and economic value while retaining some habitat and carbon, and development collapsing nearly all of them. Development is also the least reversible outcome: converted land does not return to grassland, and it takes the largest share of what is lost (Fig. 6b). Grazing carries the methane cost on the climate axis, but it is the only future that keeps the other four benefits available at once.

Discussion

Taken one axis at a time, extensive cattle grazing is easy to caricature, either as a climate liability or as a pastoral good. Assessed as a whole, and against the uses that actually replace it, peri-urban grazing emerges as a multifunctional use of land that delivers fire-fuel reduction, native biodiversity, soil carbon, and a rural economy from the same acres, at the price of a real but bounded methane cost. The central finding is not that grazing is uniformly beneficial but that its benefits are jointly produced and jointly lost: they stand or fall with the decision to keep the land grazed rather than convert it. For a community on the urban edge, the consequential choice is therefore not how many cattle to keep but whether the working landscape survives at all.

Enteric methane is a genuine cost and should not be minimized. Ruminant meat is the most emissions-intensive protein in common use, and no plausible soil-carbon credit reverses that on a per-kilogram basis[2]. The soil-carbon and waste-diversion benefits documented here are best read as partial offsets and as co-benefits of keeping land in grazed open space, not as a license to expand herds. Where they matter most is in the comparison against development, which removes the soil sink, the habitat, and the fuel management together while adding an urban emissions profile of its own.

Readers should note several limitations that bound these conclusions. The evidence for biodiversity is a grazed-versus-ungrazed contrast rather than a dose-response gradient, so it speaks to the presence of moderate grazing rather than its optimal intensity. The soil-carbon gains are strongest under compost amendment and organic management, not grazing alone, and deep-profile sampling shows that surface-only measurements can overstate them. The fire benefit is conditional on weather. And the integrated scorecard is an interpretive synthesis, not a measured index; its role is to make the joint pattern legible, not to assign precise weights. Each of these is a direction for the site-specific measurement that a full assessment of the Davis rangelands would require.

Extending beyond Davis, the broader implication is that peri-urban grazing lands are best treated as multifunctional infrastructure, valued for the bundle of services they hold in place, and that policy should focus on the margin that matters most: preventing the irreversible conversion of working rangeland. In Mediterranean-climate regions facing simultaneous pressures from fire, biodiversity loss, and urban growth, the humble grazed grassland turns out to be doing a great deal of work at once.

Methods

Scope and evidence base

Leaning on published, peer-reviewed field studies for California rangelands, this Analysis prioritizes work in the Central Valley, Sierra foothills, and coastal ranges most comparable to Yolo County. Study-system land-use figures are from the 2022 USDA Census of Agriculture county profile for Yolo County and from statewide rangeland assessments[1]. Candidate studies were identified by structured literature search and read in full; the quantitative values reported in each figure legend are drawn directly from the cited sources.

Fire-behavior representation

Estimates of fine-fuel reduction (Fig. 2a) are calibrated to the fuel-removal fractions of Lovreglio et al.[4]. The fireline-intensity curve (Fig. 2b) is a logistic function of live herbaceous fuel load that reproduces the monotone rise and the $100\,\mathrm{BTU\,ft^{-1}\,s^{-1}}$ extreme-behavior threshold reported from BEHAVE Plus modeling by Launchbaugh et al.[3]; it is illustrative of the modeled relationship rather than a refit of the original model. Weather dependence (Fig. 2c) contrasts the large rate-of-spread reduction under moderate conditions with the limited effect

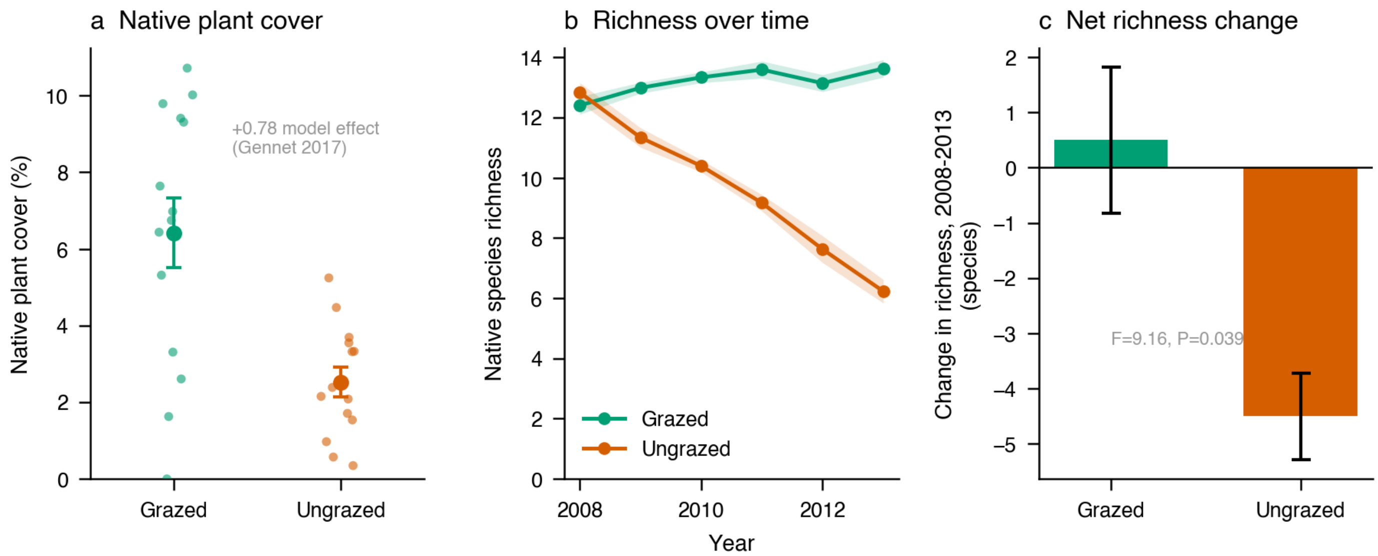

Figure 3. Grazing maintains native grassland biodiversity. **a**, Native plant cover under grazed and ungrazed conditions; points are illustrative, calibrated to the model-averaged grazing effect of Gennet et al.[5]. **b**, Native species richness over time, reproducing the divergence between grazed and ungrazed plots reported by Beck et al.[6] (mean ± 1 s.e., 6 illustrative series). **c**, Net change in richness over the study period, from Beck et al.[6] ($F = 9.16$, $P = 0.039$).

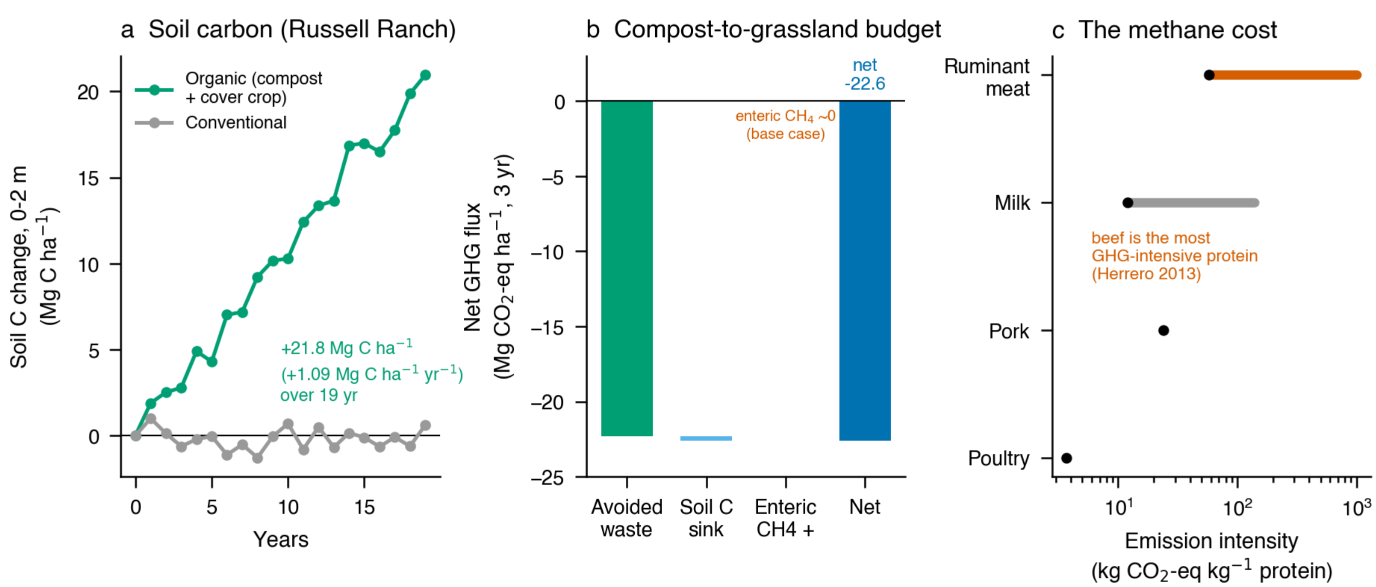

Figure 4. Soil-carbon gains versus the enteric-methane cost. **a**, Soil-carbon change under organic versus conventional management, reproducing the nineteen-year gain at Russell Ranch, Yolo County[7] (illustrative trajectory). **b**, Net greenhouse-gas budget of a single compost application to grazed grassland[8]; the benefit is dominated by avoided waste emissions. **c**, Emission intensity of protein sources, showing ruminant meat as the most intensive[2] (log scale).

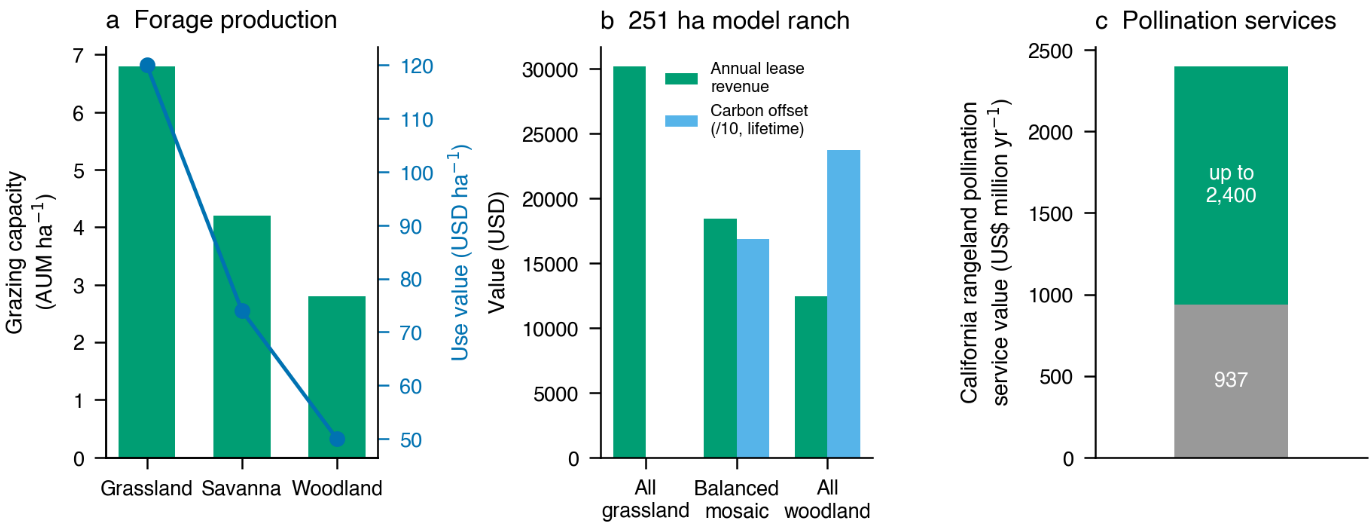

Figure 5. Forage production and the working-landscape economy. **a**, Grazing capacity and forage use value across grassland, savanna, and woodland[9]. **b**, Annual grazing revenue versus lifetime carbon-offset value for a 251-hectare model ranch under three land allocations[9] (carbon offset shown at one-tenth scale). **c**, Annual value of California rangeland pollination services[10].

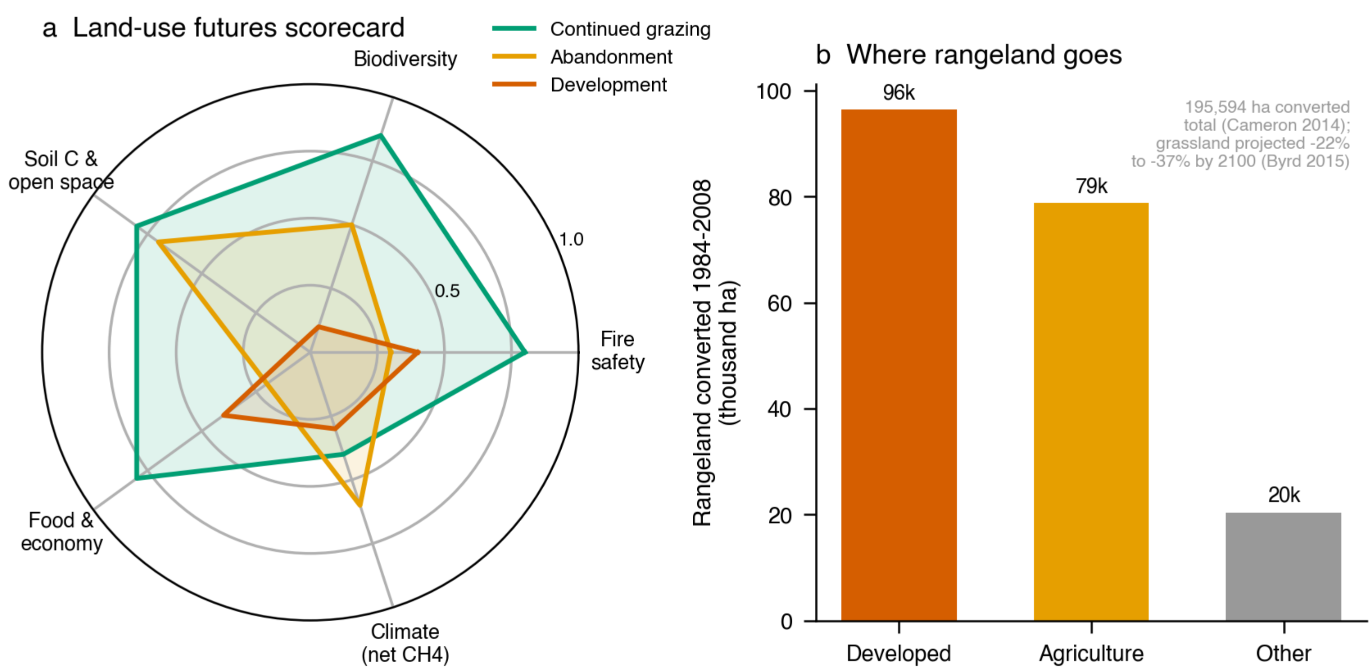

Figure 6. Three land-use futures compared. **a**, Synthesis scorecard across the five benefit dimensions for continued grazing, abandonment, and development; scores are a normalized interpretive synthesis of the evidence assembled here (see Methods), with higher indicating a better outcome. **b**, Observed rangeland conversion by destination, 1984–2008[1], with projected grassland loss by 2100[13].

under extreme fire weather reported in the same source.

Biodiversity and soil-carbon representation

Values for native cover and richness (Figs 3, 4a) are synthetic replicates for visualization, reproducing the published means, standard errors, and F-statistics of Gennet et al.[5], Beck et al.[6], and Tautges et al.[7]. The greenhouse-gas budget (Fig. 4b) plots the base-case net flux and its components from DeLonge et al.[8]; protein emission intensities (Fig. 4c) are the ranges reported by Herrero et al.[2].

Scenario synthesis

Each land-use future is assigned a normalized score between zero and one on each of the five dimensions in the scorecard (Fig. 6a), where higher denotes a better outcome for that benefit. Scores are an interpretive synthesis of the evidence assembled in this paper rather than a measured quantity: fire and biodiversity scores follow the grazing effects in Figs 2–3[3–6]; soil-carbon and open-space scores follow the retention of grazed land and its carbon under each future[7–9,13]; food and economy scores follow forage and service values[9,10]; and the climate score reflects the methane cost of grazing against the emissions of the alternatives[2,8,12]. Conversion figures (Fig. 6b) are from Cameron et al.[1] and Byrd et al.[13].

Statistics

Numerical tests reported here (F-statistics, P-values, standard errors, and model-averaged estimates) are those of the cited primary studies and are reproduced without recomputation. Synthetic replicates for visualization follow the sample sizes and dispersion of the corresponding source studies.

Data availability

Data and scripts to reproduce the figures are available upon request.

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